07/16/03

(Equatorial only)

Time Delays in GR (~~Schwarzschild~~ only)

Schwarzschild Metric ($\theta = \pi/2$):

$$ds^2 = -B(r)\,dt^2 + B^{-1}(r)\,dr^2 + r^2 d\varphi^2 ;$$

$$\boxed{B(r) = 1 - \frac{r_s}{r}} \quad ; \quad r_s = \frac{2GM}{c^2} .$$

For light, $ds^2 = 0$

$$\Rightarrow \quad 0 = -B(r)\left(\frac{dt}{d\lambda}\right)^2 + B^{-1}(r)\left(\frac{dr}{d\lambda}\right)^2 + r^2\left(\frac{d\varphi}{d\lambda}\right)^2 .$$

↳ affine parameter

Integrals of motion for t and φ:

$$E = B(r)\frac{dt}{d\lambda} \quad ; \quad L = r^2 \frac{d\varphi}{d\lambda} ;$$

$$\frac{L}{E} = r^2 B^{-1}(r)\frac{d\varphi}{dt} .$$

For $r \to \infty$, $\frac{L}{E} \to r^2 \frac{d\varphi}{dt}$ = orbital angular momentum. If the impact parameter (measured at ∞) is b, then

$$\left(r^2 \frac{d\varphi}{dt}\right)_{r\to\infty} = b \cdot \overset{=1}{c} = b = \frac{L}{E} ;$$

$$\Rightarrow \quad b = r^2 B^{-1}(r)\frac{d\varphi}{dt} ;$$ b carries sign of L, but I'll take $b > 0$ from here on.

Equation of motion:

$$0 = -1 + B^{-2}(r)\left(\frac{dr}{dt}\right)^2 + B^{-1}(r)\, r^2 \underbrace{\left(\frac{d\phi}{dt}\right)^2}_{= b^2 r^{-4} B^2(r)}$$

$$= -1 + B^{-2}(r)\left(\frac{dr}{dt}\right)^2 + B(r)\, b^2 r^{-2} = 0.$$

At close approach of the photon trajectory:

$$r = r_0;\ \frac{dr}{dt} = 0 \Rightarrow 0 = -1 + B(r_0)\, b^2 r_0^{-2}$$

$$\Rightarrow b^2 = r_0^2 B^{-1}(r_0) = r_0^2 \left(1 - \frac{r_s}{r_0}\right)^{-1}$$

$$\Rightarrow (b/r_0)^2 = 1 + \frac{r_s}{r_0} + \left(\frac{r_s}{r_0}\right)^2 + O\left[\left(\frac{r_s}{r_0}\right)^3\right].$$

So, let's write the equation of motion in a form that depends on r_0 instead of b:

$$0 = -1 + B^{-2}(r)\left(\frac{dr}{dt}\right)^2 + r^{-2} B(r)\, r_0^2 B^{-1}(r_0)$$

$$\Rightarrow \boxed{B^{-2}(r)\left(\frac{dr}{dt}\right)^2 + \frac{B(r)}{B(r_0)}\left(\frac{r_0}{r}\right)^2 - 1 = 0}.$$

Now, what is <u>elapsed</u> <u>coordinate</u> <u>time</u> for a photon traveling from $(r_1 > r_0, \phi < 0)$ to $(r_2 > r_0, \phi > 0)$, where $\phi = 0$ corresponds to closest approach?

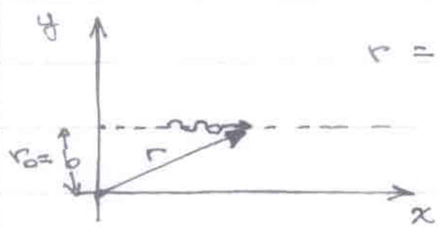
$$\left(\frac{dt}{dr}\right)^2 = B^{-2}(r)\left\{1 - \frac{B(r)}{B(r_0)}\left(\frac{r_0}{r}\right)^2\right\}^{-1}$$

$\Rightarrow$

$$t(r_0, r > r_0) = \int_{r_0}^{r} dr\, B^{-1}(r)\left\{1 - \frac{B(r)}{B(r_0)}\left(\frac{r_0}{r}\right)^2\right\}^{-1/2}.$$

Note: for $r_s \to 0$, we find

$$t(r_0, r) = \int_{r_0}^{r} dr \left[1 - \left(\frac{r_0}{r}\right)^2\right]^{-1/2} ; \quad \underbrace{b \longleftrightarrow r_0}_{\text{no bending}}$$



$$r = (r_0^2 + x^2)^{1/2} \Rightarrow \dot{r} = x \underset{1=c}{\dot{x}} \frac{1}{r} = x/r$$
$$= \left[1 - \left(\frac{r_0}{r}\right)^2\right]$$

$$\Rightarrow \frac{dt}{dr} = \left[1 - \left(\frac{r_0}{r}\right)^2\right]^{-1/2}.$$

So, the above is (obviously) for a straight-line path, with the (obvious) solution:

$$x = t ; \quad t^2 = x^2 = \underline{\underline{(r^2 - r_0^2)^{1/2}}}.$$

Therefore, it would be advantageous to [redacted] isolate the 'Pythagorean' term as a multiplier in the integrand, so that, by construction, the remaining terms contain the higher-order effects.

Begin by writing

$$1-\frac{B(r)}{B(r_0)}\left(\frac{r_0}{r}\right)^2 = 1-\left(\frac{r_0}{r}\right)^2+\left(\frac{r_0}{r}\right)^2\left[1-\frac{B(r)}{B(r_0)}\right]$$

$$\Rightarrow t(r_0,r) = \int_{r_0}^{r} dr\left[1-\left(\frac{r_0}{r}\right)^2\right]^{-1/2} B^{-1}(r)\left\{1+\left(\frac{r_0}{r}\right)^2\left[1-\left(\frac{r_0}{r}\right)^2\right]^{-1} \times\left[1-\frac{B(r)}{B(r_0)}\right]\right\}^{-1/2}$$

let $\boxed{F(r;r_0)} = B^{-1}(r)\left\{1+\left(\frac{r_0}{r}\right)^2\left[1-\left(\frac{r_0}{r}\right)^2\right]^{-1}\left[1-\frac{B(r)}{B(r_0)}\right]\right\}^{-1/2}$;

$F(r;r_0)$ all the <u>relativistic</u> <u>effects</u>.

For convenience, change variables to

$\boxed{u=\frac{r_0}{r}} \Rightarrow \frac{r_s}{r} = \frac{r_s}{r_0}\frac{r_0}{r} = \xi u$;

$\boxed{\xi = \frac{r_s}{r_0}} \rightarrow$ <u>perturbative parameter</u>

$$\Rightarrow F(u) = B^{-1}\left(r\to\frac{r_0}{u}\right)\left\{1+u^2(1-u^2)^{-1}\left[1-\frac{B(u)}{B(1)}\right]\right\}^{-1/2}$$

$\underline{B(u) = 1-\xi u}$; $\underline{B(1) = 1-\xi}$

$$\Rightarrow F(u) = (1-\xi u)^{-1}\left\{1+\frac{u^2}{1-u^2}\left[1-\frac{(1-\xi u)}{(1-\xi)}\right]\right\}^{-1/2}$$

$$\{\} = 1+\frac{u^2}{1-u^2}\left[\frac{\cancel{1}-\xi-\cancel{1}+\xi u}{1-\xi}\right] = 1+\frac{u^2}{1-u^2}\left[-\frac{\xi(1-u)}{1-\xi}\right]$$

$$= 1-\xi(1-\xi)^{-1}(1-u^2)^{-1}(1-u)u^2$$

$$= 1-\xi(1-\xi)^{-1}(1+u)^{-1}u^2 \rightarrow \text{nonsingular for } u\to 1.$$

$$\Rightarrow F(u) = (1-\xi u)^{-1}\left\{1 - \xi(1-\xi)^{-1}\frac{u^2}{1+u}\right\}^{-1/2}$$

$$(1-\xi u)^{-1} = 1 + \xi u + \xi^2 u^2 + O(\xi^3)$$
$$(1-\xi)^{-1} = \underline{\underline{1+\xi}} + \xi^2 + O(\xi^3)$$

$$\Rightarrow F(u) \simeq (1+\xi u + \xi^2 u^2)\left[1 - \xi(1+\xi)\frac{u^2}{1+u}\right]^{-1/2}$$

$$(1-x)^{-1/2} = 1 + \tfrac{1}{2}x + \tfrac{1}{2}(-\tfrac{1}{2})(-\tfrac{1}{2}-1)\,x^2 + O(x^3)$$
$$= 1 + \tfrac{1}{2}x + \tfrac{3}{8}x^2 + O(x^3)$$

$$x^2 = \xi^2(1+\xi)^2\frac{u^4}{(1+u)^2} = \xi^2\frac{u^4}{(1+u)^2} + O(\xi^3)$$

$$\Rightarrow [\;]^{-1/2} = 1 + \tfrac{1}{2}\xi(1+\xi)\frac{u^2}{1+u} + \tfrac{3}{8}\xi^2\frac{u^4}{(1+u)^2}$$
$$= 1 + \tfrac{1}{2}\xi\frac{u^2}{1+u} + \xi^2\left[\tfrac{1}{2}\frac{u^2}{1+u} + \tfrac{3}{8}\frac{u^4}{(1+u)^2}\right]$$

$$\Rightarrow F(u) \simeq [1+\xi u + \xi^2 u^2]\left[1 + \xi\cdot\tfrac{1}{2}\frac{u^2}{1+u} + \xi^2\left(\tfrac{1}{2}\frac{u^2}{1+u} + \tfrac{3}{8}\frac{u^4}{(1+u)^2}\right)\right]$$

$$\cong 1 + \xi u + \xi^2 u^2 + \xi\cdot\tfrac{1}{2}\frac{u^2}{1+u} + \xi^2\cdot\tfrac{1}{2}\frac{u^3}{1+u}$$
$$+ \xi^2\left(\tfrac{1}{2}\frac{u^2}{1+u} + \tfrac{3}{8}\frac{u^4}{(1+u)^2}\right)$$
$$= 1 + \xi\left[u + \tfrac{1}{2}\frac{u^2}{1+u}\right] + \xi^2\left[u^2 + \tfrac{1}{2}\frac{u^3}{1+u} + \tfrac{1}{2}\frac{u^2}{1+u} + \tfrac{3}{8}\frac{u^4}{(1+u)^2}\right]$$

$$= 1 + \xi\left[u + \tfrac{1}{2}\frac{u^2}{1+u}\right] + \xi^2 u^2\left[1 + \tfrac{1}{2}\frac{u}{1+u} + \tfrac{1}{2}\frac{1}{1+u} + \tfrac{3}{8}\frac{u^2}{(1+u)^2}\right]$$

$$= 1 + \xi u^2\left[\frac{1}{u} + \tfrac{1}{2}\frac{1}{1+u}\right] + \xi^2 u^2\left[1 + \tfrac{1}{2}\cancelto{1}{\frac{u+1}{1+u}} + \tfrac{3}{8}\frac{u^2}{(1+u)^2}\right]$$

$$= \boxed{1 + \xi u^2\left[\frac{1}{u} + \tfrac{1}{2}\frac{1}{1+u}\right] + \xi^2 u^2\left[\tfrac{3}{2} + \tfrac{3}{8}\frac{u^2}{(1+u)^2}\right]}$$

Now, to $O(\xi^2)$, the time integral is as follows:

$$F(u) = 1 + \xi \underbrace{u^2\left[\frac{1}{u} + \frac{1}{2}\frac{1}{1+u}\right]}_{f_1(u)} + \xi^2 \underbrace{u^2\left[\frac{3}{2} + \frac{3}{8}\frac{u^2}{(1+u)^2}\right]}_{f_2(u)} + O(\xi^3)$$

$$= 1 + \xi f_1(u) + \xi^2 f_2(u) + O(\xi^3)$$

'Pythagorean' 1st order Shapiro 2nd order Shapiro

I can drop the $\overbrace{\text{first}}^{\text{'Pythagorean'}}$ term.

The $\boxed{\text{first-order}}$ $\boxed{\text{Shapiro delay}}$ is:

$$t_1(\overset{\to u_0}{r_0}, \overset{\to u}{r}) = \underbrace{\overset{\boxed{=r_s}}{r_0 \xi} \int_u^{u_0=1}}_{dr = -\frac{r_0}{u^2}du} du\, u^{-2}(1-u^2)^{-1/2} f_1(u)$$

$$\begin{aligned}
I_1 &= \int du\, u^{-2}(1-u^2)^{-1/2} f_1(u) \\
&= \int du\, \cancel{u^{-2}}(1-u^2)^{-1/2} \cancel{u^2}\left[\frac{1}{u} + \frac{1}{2}\frac{1}{1+u}\right] \\
&= \int du\, (1-u^2)^{-1/2}\left[\frac{1}{u} + \frac{1}{2}\frac{1}{1+u}\right] \\
&= \underbrace{\int du\, (1-u^2)^{-1/2} u^{-1}}_{I_1^a} + \frac{1}{2}\underbrace{\int du\, (1-u^2)^{-1/2}(1+u)^{-1}}_{I_1^b} = \underline{\underline{I_1^a + \frac{1}{2}I_1^b}}
\end{aligned}$$

I_1^b: let $x = 1+u \Rightarrow u = (x-1)$; $u^2 = x^2 - 2x + 1$

$1 - u^2 = -x^2 + 2x$; $dx = du$

$$I_1^b = \int dx\, (0 + 2x - x^2)^{-1/2} x^{-1}$$

Both I_1^a and I_1^b are of the form $\underline{\underline{\int dy\, (a + by + cy^2)^{-1/2}}} y^{-1}$.

From Peirce + Foster, 'A Short Table of Integrals,' pg. 28

$$\varphi = a + by + cy^2$$

$$\int dy\, y^{-1} \varphi^{-1/2} = -\frac{1}{a^{1/2}} \ln\left[\frac{\varphi^{1/2} + a^{1/2}}{y} + \frac{b}{2a^{1/2}}\right] \quad , \quad \underline{\underline{a > 0}}$$

$$= \frac{-2\varphi^{1/2}}{by} \quad , \quad \underline{\underline{a = 0}}$$

Therefore,

I_1^a: $\quad a = 1, b = 0, c = -1$

$$\Rightarrow I_1^a = -\ln\left[\frac{1 + (1-u^2)^{1/2}}{u}\right]$$

I_1^b: $\quad a = 0, b = 2, c = -1$

$$\Rightarrow I_1^b = \frac{-\cancel{2}(2x - x^2)}{\cancel{2}x} = -\frac{1}{1+u}(1-u^2)^{1/2}$$

$$\Rightarrow I_1 = I_1^a + \frac{1}{2} I_1^b$$
$$= -\ln\left[\frac{1 + (1-u^2)^{1/2}}{u}\right] - \frac{1}{2}\frac{(1-u^2)^{1/2}}{1+u}$$

$$\Rightarrow t_1(u,1) = r_s \int_u^1 du\, (1-u^2)^{-1/2} f_1(u)$$
$$= r_s \left[I_1^a \Big|_u^1 + \frac{1}{2} I_1^b \Big|_u^1 \right]$$

$$I_1^a(1) = -\ln(1) = 0 \quad ; \quad I_1^b(1) = 0$$

$$\Rightarrow t_1(u,1) = r_s \left[\ln\left(\frac{1 + (1-u^2)^{1/2}}{u}\right) + \frac{1}{2}\frac{(1-u^2)^{1/2}}{1+u}\right]$$

Putting back $u = \frac{r_0}{r}$, we find

$$\frac{1 + (1-u^2)^{1/2}}{u} = \frac{1}{r_0}\left[r + (r^2 - r_0^2)^{1/2}\right]$$

$$\frac{(1-u^2)^{1/2}}{1+u} = \frac{(r^2-r_0^2)^{1/2}}{r+r_0} = \left(\frac{r-r_0}{r+r_0}\right)^{1/2}$$

$$\Rightarrow \boxed{t_1(r_0, r) = r_s \left\{ \ln\left[\frac{r + (r^2-r_0^2)^{1/2}}{r_0}\right] + \frac{1}{2}\left(\frac{r-r_0}{r+r_0}\right)^{1/2} \right\}}$$

[This is the correct first-order Shapiro delay.]

The second-order delay is as follows:

$$t_2(u,1) = \overbrace{r_0 \, \xi^2}^{r_s^2/r_0} \int_u^1 du \, (1-u^2)^{-1/2} u^{-2} f_2(u)$$

$$= \frac{r_s^2}{r_0} \int_u^1 du \, (1-u^2)^{-1/2} \left[\frac{3}{2} + \frac{3}{8}\frac{u^2}{(1+u)^2}\right]$$

$$= \frac{3}{2}\frac{r_s^2}{r_0}\left\{ \underbrace{\int_u^1 du \, (1-u^2)^{-1/2}}_{I_2^a} + \frac{1}{4}\underbrace{\int_u^1 du \, (1-u^2)^{-1/2} \frac{u^2}{(1+u)^2}}_{I_2^b} \right\}$$

$$I_2^b = \int_u^1 du \, (1-u^2)^{-1/2} \frac{u^2}{(1+u)^2}$$

Let $x = 1+u \Rightarrow u = x-1$, $u^2 = x^2 - 2x + 1$; $1 - u^2 = 2x - x^2$

$$\Rightarrow \int dx \, (2x - x^2)^{-1/2} \frac{(x^2 - 2x + 1)}{x^2}$$

$$= \int dx \, (0 + 2x - x^2)^{-1/2} \underline{\left[1 - 2x^{-1} + x^{-2}\right]}$$

From Peirce + Foster:

$Y = a + by + cy^2$;

$$\underline{\underline{\int dy\, y^{-1} Y^{-1/2}}} = -\frac{2Y^{1/2}}{by}, \quad \underline{\underline{a = 0}}$$

$$\underline{\underline{\int dy\, y^{-2} Y^{-1/2}}} = -\frac{2}{3}\frac{1}{b}\left[\frac{Y^{1/2}}{y^2} + c\int dy\, y^{-1} Y^{-1/2}\right], \quad \underline{\underline{a = 0}}$$

$$= -\frac{2}{3}\frac{1}{b}\left[\frac{Y^{1/2}}{y^2} - \frac{2cY^{1/2}}{by}\right]$$

$$= -\frac{2}{3b}(by + cy^2)^{1/2} y^{-2}\left[1 - \frac{2c}{b}y\right]$$

[redacted]

$$\underline{\underline{\int dy\, Y^{-1/2}}} = -\frac{1}{(-c)^{1/2}} \sin^{-1}\left[1 + \frac{2c}{b}y\right], \quad \underline{\underline{a = 0}}$$

So, for I_2^b we have as follows:

$$I_2^b = \underbrace{\left[\int dx\, (2x - x^2)^{-1/2}\right]}_{(i)} - 2\overbrace{\left[\int dx\, (2x - x^2)^{-1/2} x^{-1}\right]}^{(ii)} + \underbrace{\left[\int dx\, (2x - x^2)^{-1/2} x^{-2}\right]}_{(iii)}$$

Here, $a = 0$, $b = 2$, $c = -1$;

$$(i) = -\sin^{-1}\{1 - \overset{x}{\cancel{y}}\} = -\sin^{-1}(1 - x) = + \underline{\underline{\sin^{-1} u}}$$

$$(ii) = -x^{-1}(2x - x^2)^{1/2} = -\frac{1}{1+u}(1 - u^2)^{1/2} = -\underline{\underline{\left(\frac{1-u}{1+u}\right)^{1/2}}}$$

$$(iii) = -\frac{1}{3}(2x - x^2)^{1/2} x^{-2} (1+x)$$
$$= -\frac{1}{3}(1-u^2)^{\frac{1}{2}} (1+u)^{-2} (2+u)$$

$$= -\frac{1}{3}\left(\frac{2+u}{1+u}\right)\left(\frac{1-u}{1+u}\right)^{1/2}$$

So,

$$I_2^b = (i) - 2(ii) + (iii)$$
$$= \sin^{-1}u + 2\left(\frac{1-u}{1+u}\right)^{1/2} - \frac{1}{3}\left(\frac{2+u}{1+u}\right)\left(\frac{1-u}{1+u}\right)^{1/2}$$

$$= \sin^{-1}u + \frac{1}{3}\left(\frac{4+5u}{1+u}\right)\left(\frac{1-u}{1+u}\right)^{1/2}.$$

Finally,

$$I_2^b \Big|_u^1 = \left[\frac{\pi}{2} + 0\right] - \left[\sin^{-1}u + \frac{1}{3}\left(\frac{4+5u}{1+u}\right)\left(\frac{1-u}{1+u}\right)^{1/2}\right];$$

$$I_2^a = +\sin^{-1}u\Big|_u^1 = \frac{\pi}{2} - \sin^{-1}u$$

$$\Rightarrow t_2(u,1) = \frac{3}{2}\frac{r_s^2}{r_0}\left\{I_2^a\Big|_u^1 + \frac{1}{4} I_2^b\Big|_u^1\right\}$$

$$= \frac{3}{2}\frac{r_s^2}{r_0}\left\{\left(\frac{\pi}{2} - \sin^{-1}u\right) + \frac{1}{4}\left[\left(\frac{\pi}{2} - \sin^{-1}u\right) - \frac{1}{3}\left(\frac{4+5u}{1+u}\right)\left(\frac{1-u}{1+u}\right)^{1/2}\right]\right\}$$
$$= \frac{3}{2}\frac{r_s^2}{r_0}\left\{\frac{5}{4}\left(\frac{\pi}{2} - \sin^{-1}u\right) - \frac{1}{12}\left(\frac{4+5u}{1+u}\right)\left(\frac{1-u}{1+u}\right)^{1/2}\right\}$$

Second-order Shapiro delay:

$$t_2(u,1) = \frac{r_s^2}{r_0}\left\{\frac{15}{8}\left(\frac{\pi}{2} - \sin^{-1}u\right) - \frac{1}{8}\left(\frac{4+5u}{1+u}\right)\left(\frac{1-u}{1+u}\right)^{1/2}\right\}$$

For a distant observer, $u << 1$:

$$t_2(u,1) \simeq \frac{r_s^2}{r_0}\left\{\frac{15}{8}\left(\frac{\pi}{2}-u\right) - \frac{1}{8}(4-3u)\right\}$$
$$= \frac{r_s^2}{r_0}\left\{\frac{1}{8}\left[\frac{15\pi}{2} - 15u - 4 + 3u\right]\right\}$$
$$= \frac{r_s^2}{r_0}\cdot\frac{1}{8}\left\{\frac{15\pi-8}{2} - (12u)\right\}$$

$$= \underbrace{\boxed{\frac{r_s^2}{r_0}\left(\frac{15\pi-8}{16}\right)}} + O(u)$$

This expression agrees exactly with Laguna & Wolszczan (1997).